

NIHAL AKNDO

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SUMMARY

Self-taught AI/ML engineer specializing in agentic systems, computer vision, and retrieval-augmented architectures. Experienced in designing end-to-end pipelines spanning embedded systems, real-time inference, and full-stack application delivery, with a long-term focus on building applied AI products.

EDUCATION

B.S. in Computer Science and Engineering (In Progress / Applying)

Expected 2027

University Name, City, Country

TECHNICAL SKILLS

Languages: Python, C, C#, JavaScript, TypeScript, SQL

Machine Learning & AI: Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), NLP, Computer Vision, Deep Learning

Computer Vision: OpenCV, YOLO, OCR, Image Processing

Data Science: NumPy, Pandas, Matplotlib, Data Preprocessing, Feature Engineering

Backend & APIs: FastAPI, Flask, REST APIs, WebSockets

Frontend: React, HTML5, CSS3, Three.js

Databases: PostgreSQL, SQLite, MongoDB, FAISS, ChromaDB

Developer Tools: Git, GitHub, Docker, Jupyter Notebook, VS Code, Linux, Postman

Automation & Web Scraping: Playwright, Selenium, BeautifulSoup, Requests

PROJECTS

S.I.I — Agentic AI Desktop Assistant *Python, FastAPI, LangGraph, Ollama/OpenAI, FAISS, Whisper, PySide6, OpenCV*

- Architected an agentic AI desktop assistant leveraging tool-calling, retrieval-augmented generation (RAG), and persistent semantic memory via FAISS vector indexing.
- Developed asynchronous pipelines for speech recognition (Whisper), LLM inference, text-to-speech synthesis, and OS-level automation within a modular plugin architecture.

VisionOS — Real-Time Computer Vision Platform *PyTorch, OpenCV, YOLO, TensorRT, FastAPI, CUDA*

- Engineered a modular real-time computer vision framework supporting object detection, multi-object tracking, OCR, and face recognition.
- Optimized GPU inference pipelines using TensorRT and ONNX Runtime for low-latency video analytics, with an event-driven REST/WebSocket architecture for live inference streaming.

AI Research Assistant *LangChain, LangGraph, FAISS, ChromaDB, FastAPI, React*

- Built a retrieval-augmented knowledge platform for semantic document search and citation-aware question answering, with embedding generation and vector indexing pipelines for heterogeneous data sources.
- Implemented hybrid retrieval and context-ranking strategies to improve LLM response quality, exposed through modular backend services via REST APIs.

Space Intelligence Dashboard *Python, FastAPI, Three.js, PostgreSQL, WebSockets*

- Developed a real-time aerospace intelligence platform aggregating telemetry from multiple public space APIs, with interactive 3D orbital visualizations for satellite, launch, and asteroid monitoring.
- Implemented asynchronous data ingestion pipelines with automated synchronization and caching, integrating LLM-powered analytics for natural-language querying and event summarization.

Autonomous Robotics Intelligence Stack *ROS2, ESP32, OpenCV, PyTorch, MQTT, FastAPI*

- Designed a distributed robotics software stack integrating embedded hardware, computer vision, and autonomous planning, with real-time sensor fusion and perception modules for navigation.
- Developed low-latency communication between edge devices and AI inference services using MQTT, with modular control interfaces extensible to SLAM and manipulation.

COURSEWORK & CERTIFICATIONS

- Foundational CS & Programming:** CS50x, CS50P, CS50: Introduction to AI with Python (Harvard/edX); Python for Everybody (Coursera)
- ML/DL Specializations:** Machine Learning, Deep Learning, Mathematics for Machine Learning & Data Science (Andrew Ng, DeepLearning.AI)
- Advanced / Research-Level:** Stanford CS231n — Computer Vision; Stanford CS224N — NLP with Deep Learning; Neural Networks: Zero to Hero (Andrej Karpathy)
- Other:** Web Scraping with Python (Udemy)